Lecture 3

Mixtures and the Bloch ball

Objectives. *Atoms straight from an oven read* +1 *with probability* $\frac{1}{2}$ *along every axis, and no state of Lecture 1 does that. Such a beam is a random mixture of states. This lecture shows that everything a measurement can reveal about a mixture is held in one matrix, the density matrix. For a qubit, the density matrices fill the ball inside the Bloch sphere.*

3.1 An unpolarised beam

Send the silver atoms straight from the oven into a Stern–Gerlach magnet, with no magnet before it. Whatever the direction $\mathbf{m}$ of the magnet, half of the atoms read +1 and half read −1. A beam with this property is called *unpolarised*.

No state of Lecture 1 gives this. By Corollary 1.8, the state with Bloch vector $\mathbf{r}$ reads +1 along $\mathbf{m} = \mathbf{r}$ with certainty. So the atoms from the oven are not all in one state.

A natural guess is that each atom leaves the oven in some state, and that the state varies at random from atom to atom. Suppose, for instance, that the oven prepares $|\uparrow\rangle$ or $|\downarrow\rangle$, each with probability $\frac{1}{2}$. Along an axis $\mathbf{m}$, the atoms prepared $|\uparrow\rangle$ read +1 with probability $\frac{1}{2}(1 + m_z)$, and those prepared $|\downarrow\rangle$ with probability $\frac{1}{2}(1 - m_z)$. In total,

$$\tfrac{1}{2} \cdot \tfrac{1}{2}(1 + m_z) + \tfrac{1}{2} \cdot \tfrac{1}{2}(1 - m_z) = \tfrac{1}{2}$$

for every $\mathbf{m}$, as observed. But an oven that prepares $|{+x}\rangle$ or $|{-x}\rangle$, each with probability $\frac{1}{2}$, gives $\frac{1}{2}$ for every $\mathbf{m}$ just as well. Which is it? This lecture shows that no measurement can tell.

Self-check 3.1

1. A beam reads +1 along z with probability $\frac{1}{2}$ and along x with probability 1. Can it be the beam from the oven?

The physics

In an oven at several hundred degrees, silver atoms collide with the walls and with each other at random, and nothing in the oven picks out a direction in space. A sample of water in an NMR magnet, left in thermal equilibrium with its surroundings at the temperature T, is nearly unpolarised too. Statistical physics then weights the two energy states of a proton in proportion to $e^{-E/kT}$, where k is Boltzmann's constant. In a field of 1 T at 300 K this gives a Bloch vector of length $\tanh(\hbar\gamma B/2kT) \approx 3.4 \cdot 10^{-6}$. The NMR signal of Lecture 2 comes from turning this small excess into the xy-plane with a $\pi/2$-pulse.

3.2 Random preparations

The guess uses only finite probability, which we recall. A random experiment has finitely many outcomes k, with probabilities $p_k > 0$ that sum to 1. Outcomes that cannot occur are left out. The *conditional probability* $\mathrm{Prob}(b \mid k)$ is the probability of an event b when k is known to have occurred. The *law of total probability* is

$$\mathrm{Prob}(b) = \sum_k p_k \,\mathrm{Prob}(b \mid k). \tag{3.1}$$

When $\mathrm{Prob}(b) > 0$, *Bayes' rule* gives the probability of k once b is known:

$$\mathrm{Prob}(k \mid b) = \frac{p_k \,\mathrm{Prob}(b \mid k)}{\mathrm{Prob}(b)}. \tag{3.2}$$

The probability of two events in turn is the product of the probability of the first and the conditional probability of the second.

Definition 3.1 (Ensemble)**.** An *ensemble* is a finite list $\{(p_k, \psi_k)\}$ of probabilities $p_k > 0$ with $\sum_k p_k = 1$ and unit vectors $\psi_k \in \mathbb{C}^2$. It describes a preparation that gives each system, independently of the others, the state $[\psi_k]$ with probability p_k.

A single state $[\psi]$ is the ensemble with one member, $\{(1, \psi)\}$. The definitions and the branch calculation of § 3.3 work unchanged in $\mathbb{C}^n$, and Lecture 4 reuses them there.

Self-check 3.2

1. An ensemble has $|\uparrow\rangle$ with probability $\frac{1}{4}$ and $|\downarrow\rangle$ with probability $\frac{3}{4}$. Find the probability of $+1$ along x, and, given that reading, the probability that the atom was prepared $|\uparrow\rangle$.

3.3 The density matrix

An experiment on a prepared system is a sequence of steps. Between readings the state evolves by a unitary matrix, $e^{-\mathrm{i}tH/\hbar}$ of Lecture 1 or $U(t, s)$ of Lecture 2. A reading is an ideal measurement, and a *branch* of the experiment is one particular sequence of readings.

Lemma 3.2 (A branch for one state)**.** *A system in the state $[\psi]$ undergoes, in turn, a unitary U_1, a measurement that reads a_1, …, a unitary U_m, and a measurement that reads a_m. Let E_j be the projection of the reading a_j, and put*

$$K = E_m U_m \cdots E_1 U_1, \tag{3.3}$$

the branch operator. *Each U_j may depend on the earlier readings. Then the readings $a_1, \ldots, a_m$ occur with probability $\|K\psi\|^2$. If this is positive, the state afterwards is the ray of $K\psi/\|K\psi\|$.*

Proof. We use induction on m. For $m = 0$, $K = \mathbb{1}$, and there is nothing to prove. Let $K' = E_{m-1}U_{m-1}\cdots E_1U_1$, so that $K = E_mU_mK'$, and assume the lemma for K'. The matrix E_mU_m does not increase norms, so $\|K\psi\| \le \|K'\psi\|$. If $\|K'\psi\| = 0$, the first $m-1$ readings never occur, and $\|K\psi\| = 0$ as well. Otherwise, after them the state is $\chi = K'\psi/\|K'\psi\|$. It evolves to $U_m\chi$, and by Born's rule the last reading has the conditional probability

$$\|E_mU_m\chi\|^2 = \frac{\|K\psi\|^2}{\|K'\psi\|^2}.$$

The product with $\|K'\psi\|^2$ is $\|K\psi\|^2$. If $\|K\psi\| > 0$, by Lüders' rule the state afterwards is the ray of $E_m U_m \chi$, which is the ray of $K\psi$. ∎

Definition 3.3 (Density matrix). The *density matrix* of an ensemble $\{(p_k, \psi_k)\}$ is

$$\rho = \sum_k p_k \, |\psi_k\rangle\langle\psi_k| \,. \tag{3.4}$$

It is *pure* if $\rho = |\psi\rangle\langle\psi|$ for a unit vector ψ, and *mixed* otherwise.

The density matrix is self-adjoint, and its trace is $\sum_k p_k = 1$. It is *positive*: for every vector v, $\langle v, \rho v\rangle = \sum_k p_k \, |\langle\psi_k|v\rangle|^2 \geq 0$. A phase does not change $|\psi_k\rangle\langle\psi_k|$, so ρ depends only on the states of the members. We need two facts about the trace. For vectors u, w and a matrix A, $\operatorname{Tr}|u\rangle\langle w| = \langle w|u\rangle$, and $A\,|\psi\rangle\langle\psi|\,A^\dagger = |A\psi\rangle\langle A\psi|$.

Theorem 3.4 (Statistics depend on the density matrix only). *Let ρ be the density matrix of an ensemble $\{(p_k, \psi_k)\}$, and K a branch operator (3.3). The branch occurs with probability*

$$\operatorname{Prob}(K) = \operatorname{Tr}\big(K\rho K^\dagger\big). \tag{3.5}$$

If $\operatorname{Prob}(K) > 0$, the systems that took the branch form an ensemble with density matrix $K\rho K^\dagger/\operatorname{Prob}(K)$. Hence two ensembles with the same density matrix give the same probabilities in every experiment, and after every branch of positive probability they again have equal density matrices.

Proof. By Lemma 3.2, a system prepared in $[\psi_k]$ takes the branch with probability $\|K\psi_k\|^2 = \operatorname{Tr}|K\psi_k\rangle\langle K\psi_k| = \operatorname{Tr}\big(K\,|\psi_k\rangle\langle\psi_k|\,K^\dagger\big)$. By the law of total probability (3.1) and the linearity of the trace,

$$\operatorname{Prob}(K) = \sum_k p_k \operatorname{Tr}\big(K\,|\psi_k\rangle\langle\psi_k|\,K^\dagger\big) = \operatorname{Tr}\big(K\rho K^\dagger\big).$$

Next, suppose $\operatorname{Prob}(K) > 0$. By Bayes' rule (3.2), a system that took the branch was prepared in $[\psi_k]$ with probability $q_k = p_k \, \|K\psi_k\|^2 / \operatorname{Prob}(K)$. By Lemma 3.2 it is then in the state $K\psi_k/\|K\psi_k\|$. The members with $K\psi_k = 0$ have $q_k = 0$ and are dropped. The density matrix of the new ensemble is

$$\sum_k q_k \, \frac{|K\psi_k\rangle\langle K\psi_k|}{\|K\psi_k\|^2} = \frac{1}{\operatorname{Prob}(K)} \sum_k p_k \, K\,|\psi_k\rangle\langle\psi_k|\,K^\dagger = \frac{K\rho K^\dagger}{\operatorname{Prob}(K)}.$$

Both results depend on the ensemble through ρ only. ∎

The simplest branches give the rules of Lecture 1 for a density matrix. One measurement, $K = E_a$, reads a with probability $\operatorname{Tr}(E_a \rho E_a) = \operatorname{Tr}(\rho E_a)$, by the cyclicity of the trace and $E_a^2 = E_a$. If this is positive, it leaves $E_a \rho E_a / \operatorname{Tr}(\rho E_a)$. An evolution alone, $K = U$, takes ρ to $U\rho U^\dagger$. The *mean* of the readings of an observable $A = \sum_a a\, E_a$ is

$$\sum_a a \, \operatorname{Tr}(\rho E_a) = \operatorname{Tr}(\rho A). \tag{3.6}$$

Remark 3.5 (Conditioning changes the weights). After a branch, the ensemble has the weights q_k of Bayes' rule, not the old weights p_k. Averaging the members' new states with the old weights gives a different matrix in general, and wrong predictions. A member that cannot take the branch has no state afterwards at all. On a qubit the error does not show in a single reading along an axis, since every member that can give the reading is left in the same pure state. In $\mathbb{C}^n$ it shows (Lecture 4).

Self-check 3.3

1. Find the density matrix of the ensemble $\{(\frac{1}{2}, |\uparrow\rangle), (\frac{1}{2}, |{+}x\rangle)\}$.
2. Use it to find the probability of +1 along z.

3.4 The Bloch ball

For a qubit, the *Bloch vector* of a density matrix ρ is $\mathbf{r} = (\operatorname{Tr}\rho\sigma_x, \operatorname{Tr}\rho\sigma_y, \operatorname{Tr}\rho\sigma_z)$. For a pure $\rho = |\psi\rangle\langle\psi|$ it is the Bloch vector $\langle\psi, \sigma_a\psi\rangle$ of Definition 1.2, since $\operatorname{Tr}(|\psi\rangle\langle\psi|\,\sigma_a) = \langle\psi, \sigma_a\psi\rangle$.

Theorem 3.6 (The Bloch ball)**.** *Let ρ be the density matrix of an ensemble $\{(p_k, \psi_k)\}$ of qubit states, with Bloch vector $\mathbf{r}$.*

1. *$\rho = \frac{1}{2}(\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma})$, its eigenvalues are $\frac{1}{2}(1 \pm |\mathbf{r}|)$, and $|\mathbf{r}| \le 1$.*
2. *$\mathbf{r} = \sum_k p_k\, \mathbf{r}(\psi_k)$, the weighted mean of the members' Bloch vectors.*
3. *$\operatorname{Tr}\rho^2 = \frac{1}{2}(1 + |\mathbf{r}|^2)$. Moreover, $|\mathbf{r}| = 1$ if and only if $\rho^2 = \rho$, if and only if all the members are the same state.*
4. *Every $\mathbf{r} \in \mathbb{R}^3$ with $|\mathbf{r}| \le 1$ is the Bloch vector of an ensemble.*

So the density matrices of a qubit are the points of the unit ball, the Bloch ball*. The pure ones form its surface, the Bloch sphere.*

Proof.

1. The matrix ρ is self-adjoint with trace 1. By the proof of Proposition 1.7 it is $a\mathbb{1} + \mathbf{b}\cdot\boldsymbol{\sigma}$, and the traces of the Pauli matrices give $1 = \operatorname{Tr}\rho = 2a$ and $r_a = \operatorname{Tr}\rho\sigma_a = 2b_a$. So $\rho = \frac{1}{2}(\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma})$. By Proposition 1.7 its eigenvalues are $\frac{1}{2} \pm \frac{1}{2}|\mathbf{r}|$. A positive matrix has no negative eigenvalue, so $|\mathbf{r}| \le 1$.
2. By the linearity of the trace, $\operatorname{Tr}\rho\sigma_a = \sum_k p_k \operatorname{Tr}(|\psi_k\rangle\langle\psi_k|\,\sigma_a) = \sum_k p_k \langle\psi_k, \sigma_a\psi_k\rangle$.
3. By the Pauli rule, $(\mathbf{r}\cdot\boldsymbol{\sigma})^2 = |\mathbf{r}|^2\,\mathbb{1}$, so

$$\rho^2 - \rho = \tfrac{1}{4}\big(|\mathbf{r}|^2 - 1\big)\mathbb{1},$$

   and the trace of ρ^2 follows. So $\rho^2 = \rho$ exactly when $|\mathbf{r}| = 1$. Next, write $\mathbf{r}_k = \mathbf{r}(\psi_k)$, unit vectors by Theorem 1.3. By the second part,

$$|\mathbf{r}|^2 = \sum_{j,k} p_j p_k\, \mathbf{r}_j\cdot\mathbf{r}_k \le \sum_{j,k} p_j p_k = 1,$$

   with equality exactly when $\mathbf{r}_j\cdot\mathbf{r}_k = 1$ for all j, k, that is, when all the $\mathbf{r}_k$ are equal. By Theorem 1.3, equal Bloch vectors are the same state.
4. If $\mathbf{r} = 0$, take $|\uparrow\rangle$ and $|\downarrow\rangle$ with weights $\frac{1}{2}$: their density matrix is $\frac{1}{2}\mathbb{1}$. If $\mathbf{r} \ne 0$, put $\mathbf{m} = \mathbf{r}/|\mathbf{r}|$, and let $\psi_\pm$ be the states with Bloch vectors $\pm\mathbf{m}$, with projections $E_\pm = \frac{1}{2}(\mathbb{1} \pm \mathbf{m}\cdot\boldsymbol{\sigma})$ (Proposition 1.7). Then

$$\tfrac{1}{2}(1 + |\mathbf{r}|)E_+ + \tfrac{1}{2}(1 - |\mathbf{r}|)E_- = \tfrac{1}{2}\big(\mathbb{1} + |\mathbf{r}|\;\mathbf{m}\cdot\boldsymbol{\sigma}\big) = \tfrac{1}{2}(\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma}).$$

   If $|\mathbf{r}| = 1$, the second weight is 0, and that member is dropped. ∎

The rules of Lecture 1 carry over to the ball. By the first part and the traces of the Pauli matrices, a measurement along $\mathbf{m}$ reads ±1 with probability $\operatorname{Tr}(\rho E_\pm) = \frac{1}{2}(1 \pm \mathbf{m}\cdot\mathbf{r})$, and the mean of $a\mathbb{1} + b\,\mathbf{m}\cdot\boldsymbol{\sigma}$ is $a + b\,\mathbf{m}\cdot\mathbf{r}$. After the reading +1, the density matrix is E_+, the pure state at $\mathbf{m}$, whatever ρ was: for the projection $E_+ = |\chi\rangle\langle\chi|$, $E_+\rho E_+ = \langle\chi, \rho\chi\rangle\; E_+ = \operatorname{Tr}(\rho E_+)\, E_+$.

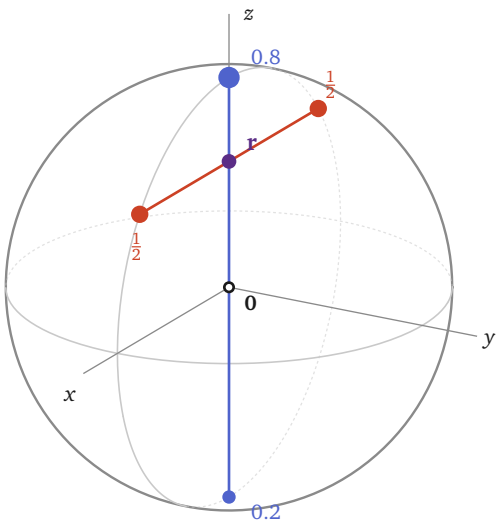


Figure 3.1: Two ensembles with one density matrix (Example 3.7). Blue: $|\uparrow\rangle$ and $|\downarrow\rangle$ with weights 0.8 and 0.2. Red: the states at $(\pm 0.8, 0, 0.6)$ with weights $\frac{1}{2}$. Both have the centre of mass $\mathbf{r} = (0, 0, 0.6)$. The centre $\mathbf{0}$ of the ball is the beam from the oven.

The same holds for -1. Under a Hamiltonian, every member's Bloch vector turns about $\mathbf{n}$ (Theorem 1.12), and by the second part so does their mean. The whole ball turns rigidly.

Self-check 3.4

1. Find the Bloch vector, the eigenvalues and $\operatorname{Tr}\rho^2$ of the density matrix in Self-check 3.3.
2. Give an ensemble of two orthogonal states with the same density matrix.

3.5 A worked example

Example 3.7 (An imperfect filter). Figure 3.1 shows the two ensembles of this example.

1. *Prepare.* A filter passes atoms in the state $|\uparrow\rangle$, but 20% of the atoms it passes are $|\downarrow\rangle$. The density matrix is

$$\rho = 0.8\,|\uparrow\rangle\langle\uparrow| + 0.2\,|\downarrow\rangle\langle\downarrow| = \begin{pmatrix} 0.8 & 0 \\ 0 & 0.2 \end{pmatrix}, \qquad \mathbf{r} = (0, 0, 0.6), \qquad \operatorname{Tr}\rho^2 = 0.68.$$

   The means are $\operatorname{Tr}\rho\sigma_z = 0.6$ and $\operatorname{Tr}\rho\sigma_x = \operatorname{Tr}\rho\sigma_y = 0$.
2. *Another ensemble.* The states $\psi_\pm = \frac{1}{\sqrt{5}}\big(2\,|\uparrow\rangle \pm |\downarrow\rangle\big)$ have the Bloch vectors $(\pm 0.8, 0, 0.6)$. With weights $\frac{1}{2}$ each, the off-diagonal entries $\pm\frac{2}{5}$ cancel, and the density matrix is the same ρ. By Theorem 3.4, no experiment tells the two ensembles apart. The weights $\frac{1}{2}$ are not the eigenvalues 0.8 and 0.2, and $\langle\psi_+|\psi_-\rangle = \frac{3}{5}$, so the states are not orthogonal.
3. *Evolve.* Under $H = \frac{\hbar\nu}{2}\sigma_y$ the ball turns about $\mathbf{e}_y$, and $\mathbf{r}(t) = 0.6\big(\sin(\nu t), 0, \cos(\nu t)\big)$. After a quarter turn, $t = \pi/(2\nu)$,

$$\mathbf{r} = (0.6, 0, 0), \qquad \rho = \begin{pmatrix} \frac{1}{2} & \frac{3}{10} \\ \frac{3}{10} & \frac{1}{2} \end{pmatrix}.$$

4. *Measure along x, then along z.* The reading +1 along x has probability $\frac{1}{2}(1+0.6) = 0.8$, and leaves the pure state $|{+}x\rangle$. A reading along z made at once then gives ± 1 with probability $\frac{1}{2}$ each. So "+1 along x, then +1 along z" has probability $0.8 \cdot \frac{1}{2} = 0.4$.
5. *Member by member.* In the first ensemble the quarter turn takes $|\uparrow\rangle$ to $|{+}x\rangle$, which reads +1 along x with certainty, and $|\downarrow\rangle$ to $|{-}x\rangle$, which never does. By Bayes' rule, an atom that read +1 was prepared $|\uparrow\rangle$ with probability $0.8 \cdot 1/0.8 = 1$. In the second ensemble the members turn to $(0.6, 0, \mp 0.8)$, and each reads +1 with probability 0.8. An atom that read +1 was prepared ψ_+ or ψ_- with probability $\frac{1}{2}$ each. The two ensembles disagree about the past of the atom, and agree on every prediction: after the reading both are the pure state $|{+}x\rangle$.

Self-check 3.5

1. In Example 3.7, what is the probability of +1 along z right after the quarter turn?
2. Suppose the reading along x is -1. What is the state afterwards, and how was the atom prepared, in the first ensemble?

3.6 What a mixture does not say

The beam from the oven has $\mathbf{r} = 0$ and $\rho = \frac{1}{2}\mathbb{1}$. The equal mixture of $|\uparrow\rangle$ and $|\downarrow\rangle$ has this density matrix, and so has the equal mixture of $|{+}x\rangle$ and $|{-}x\rangle$, or of the two states at $\pm\mathbf{m}$ for any axis $\mathbf{m}$. By Theorem 3.4, no experiment on the atoms tells these preparations apart. This answers the question of § 3.1: a density matrix does not record which states were mixed, and nothing measured on the atoms does either.

A mixture is also not a superposition. The state $|{+}x\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ and the mixture $\frac{1}{2}\mathbb{1}$ of $|\uparrow\rangle$ and $|\downarrow\rangle$ give the same statistics along z. Along x, the state $|{+}x\rangle$ reads +1 with certainty, and the mixture with probability $\frac{1}{2}$.

In this course every density matrix comes from an ensemble, or, in Lecture 11, from one half of a pair of systems. States of a particle, which need infinitely many dimensions, are not covered. *Quantum Mechanics III: Foundations* asks what a state is in general, and finds the density matrices again.

Self-check 3.6

1. Can any experiment distinguish the equal mixture of $|{+}y\rangle$ and $|{-}y\rangle$ from the beam from the oven?

The qubit, with mixtures

States: the density matrices $\rho = \frac{1}{2}(\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma})$, the points $|\mathbf{r}| \le 1$ of the Bloch ball; the pure states are its surface (Theorem 3.6). *Observables:* as in Lecture 1; the mean of A is $\mathrm{Tr}(\rho A)$. *Evolution:* $\rho \mapsto U\rho U^\dagger$; a Hamiltonian turns the ball rigidly. *Measurement:* along $\mathbf{m}$, the reading ± 1 has probability $\frac{1}{2}(1 \pm \mathbf{m}\cdot\mathbf{r})$, and the state afterwards is $\pm\mathbf{m}$. Every prediction depends on the preparation through ρ only (Theorem 3.4).

Summary.

- *Statistics depend on the density matrix only*: a branch has probability $\mathrm{Tr}(K\rho K^\dagger)$ and, if this is positive, leaves $K\rho K^\dagger$, normalised (Theorem 3.4).
- *The Bloch ball*: the density matrices of a qubit are the points of the unit ball, and the pure ones its surface (Theorem 3.6).

Lecture 4 replaces $\mathbb{C}^2$ by $\mathbb{C}^n$, keeping the postulates and the density matrix.

Self-check answers.

3.1 1. No. The oven gives $\frac{1}{2}$ along every axis. This beam is in the state $|{+x}\rangle$.

3.2 1. $\frac{1}{4}\cdot\frac{1}{2}+\frac{3}{4}\cdot\frac{1}{2}=\frac{1}{2}$, and $\mathrm{Prob}(\uparrow|\,{+1})=\frac{1}{4}\cdot\frac{1}{2}/\frac{1}{2}=\frac{1}{4}$.

3.3 1. $\rho=\frac{1}{2}\,|\uparrow\rangle\langle\uparrow|+\frac{1}{2}\,|{+x}\rangle\langle{+x}|$, with rows $(\frac{3}{4},\frac{1}{4})$ and $(\frac{1}{4},\frac{1}{4})$. 2. $\mathrm{Tr}(\rho\,|\uparrow\rangle\langle\uparrow|)=\frac{3}{4}$.

3.4 1. $\mathbf{r}=(\frac{1}{2},0,\frac{1}{2})$, eigenvalues $\frac{1}{2}\big(1\pm\frac{1}{\sqrt{2}}\big)$, $\mathrm{Tr}\,\rho^2=\frac{3}{4}$. 2. The states at $\pm(\mathbf{e}_x+\mathbf{e}_z)/\sqrt{2}$, with weights $\frac{1}{2}\big(1\pm\frac{1}{\sqrt{2}}\big)$.

3.5 1. $\frac{1}{2}$, since $r_z=0$. 2. The state $|{-x}\rangle$; the atom was prepared $|\downarrow\rangle$, with probability 1.

3.6 1. No. Both have $\rho=\frac{1}{2}\mathbb{1}$.

References

Sakurai and Napolitano, *Modern Quantum Mechanics*, §3.4, pp. 169–175. Nielsen and Chuang, *Quantum Computation and Quantum Information*, §2.4.1, pp. 99–101, Thm 2.6, p. 103, and Ex. 2.72, p. 105. Cohen-Tannoudji, Diu and Laloë, *Quantum Mechanics*, vol. 1, Complement $\mathrm{E}_{\mathrm{III}}$, pp. 299–308. Heinosaari and Ziman, *The Mathematical Language of Quantum Theory*, Ex. 2.21, pp. 62–63.